

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Monday, 11 May 2026, 5:33 PM
State	Finished
Completed on	Monday, 11 May 2026, 6:22 PM
Time taken	48 mins 30 secs
Name	BHUVANA S 2025-BME

Question **1**

Correct

Marked out of 100

Flag question

Problem Statement:

Implement an **Array-Based Linked List** (Static Linked List) using arrays to simulate linked list operations.

You are required to perform the following operations:

1. **Insert at Beginning**
2. **Insert at Position**
3. **Delete from Position**
4. **Display the List**
5. **Search an Element**
6. **Exit**

Input Format:

- The first input represents the **choice of operation**.
- Based on the choice, additional inputs may follow:
 - For insertion: position and element
 - For deletion: position
 - For search: element
- Continue performing operations until the user selects **Exit (choice 6)**.

Operations Menu:

Choice Operation

- | | |
|---|----------------|
| 1 | Insert Element |
| 2 | Delete Element |
| 3 | Search Element |
| 4 | Display List |
| 5 | Exit |
| 6 | Final Display |

Sample Input:

```
1
0
10
4
1
0
20
4
1
1
25
4
2
1
4
3
10
3
25
5
4
6
```

Sample Output:

```
10
20 10
20 25 10
20 10
```

For example:

Input	Result
1	10
0	20 10
10	20 25 10
4	20 10
1	Successful. Element 10 is at location 1
0	Unsuccessful.
20	10 20
4	
1	
1	
25	
4	
2	
1	
4	
3	
10	
3	
25	
5	
4	
6	

Answer: (penalty regime: 0 %)

Reset answer

```

1 #include <stdio.h>
2
3 int n = 0;
4
5 void Insert(int a[], int p, int e);
6 void Delete(int a[], int p);
7 void Search(int a[], int e);
8 void Traverse(int a[]);
9 void Sort(int a[]);
10
11 int main()
12 {
13     int a[5], ch, e, p;
14     do
15     {
16         scanf("%d", &ch);
17         switch(ch)
18         {
19             case 1:
20                 scanf("%d", &p);
21                 scanf("%d", &e);
22                 Insert(a,p,e);

```

	Input	Expected	Got
✓	1	10	10
	0	20 10	20 10
	10	20 25 10	20 25 10
	4	20 10	20 10
	1	Successful. Element 10 is at location 1	Successful. Element 10 is at location 1
	0	Unsuccessful.	Unsuccessful.
	20	10 20	10 20
	4		
	1		
	1		
	25		
	4		
	2		
	1		
	4		
	3		
	10		
	3		
	25		
	5		
	4		
	6		

Passed all tests! ✓

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Monday, 11 May 2026, 6:25 PM
State	Finished
Completed on	Monday, 11 May 2026, 7:05 PM
Time taken	40 mins 40 secs
Name	BHUVANA S 2025-BME

Question **1**

Correct

Marked out of
1.00Flag
question

Develop a **menu-driven program** to implement a **Singly Linked List** with the following operations:

1. **Insert at Beginning**
2. **Insert After a Given Element**
3. **Insert at End**
4. **Delete from Beginning**
5. **Delete from End**
6. **Delete a Specific Element**
7. **Search an Element**
8. **Display the List**
9. **Exit**

Input Format:

- The first input represents the **operation choice**.
- Based on the selected operation, additional inputs are taken:
 - Insert → element value
 - Insert after → existing element + new element
 - Delete specific → element
 - Search → element
- Continue operations until the user selects **Exit (9)**.

Sample Input:

```
1
40
1
30
1
20
1
10
8
7
30
1
5
8
3
45
8
2
20
25
8
4
8
6
8
5
30
8
9
```

Sample Output:

```
10 20 30 40
Element found...!
5 10 20 30 40
5 10 20 30 40 45
5 10 20 25 30 40 45
5
10 20 25 30 40 45
45
10 20 25 30 40
30
10 20 25 40
```

For example:

Input	Result
-------	--------

```

30 5 10 20 30 40 45
1 5 10 20 25 30 40 45
20 5
1 10 20 25 30 40 45
10 45
8 10 20 25 30 40
7 30
30 10 20 25 40
1
5
8
3
45
8
2
20
25
8
4
8
6
8
5
30
8
9

```

Answer: (penalty regime: 0 %)

Reset answer

```

1 /* Implementation of singly linked list - SLL.C */
2
3 #include <stdio.h>
4 #include <stdlib.h>
5
6 struct node
7 {
8     int Element;
9     struct node *Next;
10 };
11 typedef struct node Node;
12
13 int IsEmpty(Node *List);
14 int IsLast(Node *Position);
15
16 Node *Find(Node *List, int x);
17 Node *FindPrevious(Node *List, int x);
18 Node *FindNext(Node *List, int x);
19 void InsertBeg(Node *List, int e);
20 void InsertLast(Node *List, int e);
21 void InsertMid(Node *List, int p, int e);
22 void DeleteBeg(Node *List);

```

	Input	Expected	Got	
✓	1	10 20 30 40	10 20 30 40	✓
	40	Element found...!	Element found...!	
	1	5 10 20 30 40	5 10 20 30 40	
	30	5 10 20 30 40 45	5 10 20 30 40 45	
	1	5 10 20 25 30 40 45	5 10 20 25 30 40 45	
	20	5	5	
	1	10 20 25 30 40 45	10 20 25 30 40 45	
	10	45	45	
	8	10 20 25 30 40	10 20 25 30 40	
	7	30	30	
	30	10 20 25 40	10 20 25 40	
	1			
	5			
	8			
	3			
	45			
	8			
	2			
	20			
	25			
	8			
	4			
	8			
	6			
	8			
	5			
	30			
	8			
	9			

Passed all tests! ✓